

Dog Health Monitoring Dashboard - Arduino/ESP32 Code

Formatted source code export for GitHub upload/documentation.

```
#include <WiFi.h>
#include <Wire.h>

const char* ssid = "Net eskoo saab";
const char* password = "p.nitesh";

WiFiServer server(80);

int16_t ax, ay, az;
long movement;

float temperature;
int energy;
int sleepQuality;

String status="Sleeping";
String prediction="Dog resting normally";

const int maxRows = 10;

long movementHistory[maxRows];
String statusHistory[maxRows];
int timeHistory[maxRows];

int indexPos = 0;

void setup(){
  Serial.begin(115200);
  Wire.begin(21,22);

  Wire.beginTransaction(0x68);
  Wire.write(0x6B);
  Wire.write(0);
  Wire.endTransmission(true);

  WiFi.begin(ssid,password);

  Serial.print("Connecting to WiFi");

  while(WiFi.status()!=WL_CONNECTED){
    delay(500);
    Serial.print(".");
  }

  Serial.println("");
  Serial.print("Open browser: ");
  Serial.println(WiFi.localIP());

  server.begin();
}

void loop(){
  Wire.beginTransaction(0x68);
  Wire.write(0x3B);
  Wire.endTransmission(false);
  Wire.requestFrom(0x68,6,true);

  ax = Wire.read()<<8 | Wire.read();
  ay = Wire.read()<<8 | Wire.read();
  az = Wire.read()<<8 | Wire.read();

  movement = abs(ax)+abs(ay)+abs(az);

  temperature = 37.0 + random(-5,5)*0.1;
  energy = random(60,95);
  sleepQuality = random(70,100);

  if(movement < 21000){
    status="Sleeping";
    prediction="Dog resting well";
  }

  else if(movement < 45000){
    status="Walking";
    prediction="Healthy activity";
  }
  else{
```

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status="Running";
prediction="High activity detected";
}

movementHistory[indexPos] = movement;
statusHistory[indexPos] = status;
timeHistory[indexPos] = millis()/1000;

indexPos++;

if(indexPos >= maxRows){
indexPos = 0;
}

WiFiClient client = server.available();

if(client){

client.readStringUntil('\r');

client.println("HTTP/1.1 200 OK");
client.println("Content-type:text/html");
client.println();

client.println("<html>");

client.println("<head>");
client.println("<meta http-equiv='refresh' content='2'>");

client.println("<script src='https://cdn.jsdelivr.net/npm/chart.js'></script>");

client.println("<style>");

client.println("body{font-family:Arial;background:#f4f6f9;margin:0;padding:0;}");

client.println(".app{width:1100px;margin:auto;padding:20px;}");
client.println(".grid{display:grid;grid-template-columns:1fr 1fr;grid-gap:20px;}");
client.println(".card{background:white;border-radius:12px;padding:20px;box-shadow:0 4px 15px rgba(0,0,0,0.08)}");

client.println(".metric{font-size:22px;font-weight:bold;}");
client.println("table{width:100%;border-collapse:collapse;}");
client.println("th{background:#1f2937;color:white;padding:10px;}");
client.println("td{padding:10px;border-bottom:1px solid #eee;}");
client.println(".predict{background:#f8fafc;border-left:6px solid #2ECC71;padding:15px;}");
client.println("</style>");
client.println("</head>");
client.println("<body>");

client.println("<div class='app'>");
client.println("<h1>Dog Health Monitoring Dashboard</h1>");
client.println("<div class='grid'>

// DOG STATUS PANEL

client.println("<div class='card'>");
client.println("<h3>Dog Status</h3>");

if(status=="Sleeping")
client.println("<img src='https://raw.githubusercontent.com/RatnaShamith/Animal-Behaviour-detection-system/561")
else if(status=="Walking")
client.println("<img src='https://raw.githubusercontent.com/RatnaShamith/Animal-Behaviour-detection-system/0e")
else
client.println("<img src='https://raw.githubusercontent.com/RatnaShamith/Animal-Behaviour-detection-system/0e")
client.println("<p class='metric'>"+status+"</p>");

client.println("</div>");
// HEALTH METRICS
client.println("<div class='card'>");

```

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client.println("<h3>Health Metrics</h3>");

client.print("<p>Body Temperature: ");
client.print(temperature);
client.println(" °C</p>");

client.print("<p>Energy Level: ");
client.print(energy);
client.println("%</p>");

client.print("<p>Sleep Quality: ");
client.print(sleepQuality);
client.println("%</p>");

client.println("</div>");

// PIE CHART

client.println("<div class='card'>");

client.println("<h3>Activity Distribution</h3>");

client.println("<canvas id='chart'></canvas>");

client.println("</div>");

// TABLE

client.println("<div class='card'>");

client.println("<h3>Activity Log</h3>");

client.println("<table>");

client.println("<tr><th>Time</th><th>Movement</th><th>Status</th></tr>");

for(int i=0;i<maxRows;i++){

client.print("<tr><td>");
client.print(timeHistory[i]);
client.print("</td><td>");
client.print(movementHistory[i]);
client.print("</td><td>");
client.print(statusHistory[i]);
client.println("</td></tr>");

}

client.println("</table>");

client.println("</div>");

// PREDICTION

client.println("<div class='card predict'>");

client.println("<h3>AI Prediction</h3>");

client.println("<p>"+prediction+"</p>");

client.println("<p>Recommended walk: 30 minutes daily</p>");

client.println("</div>");

client.println("</div>");

// CHART SCRIPT

client.println("<script>");

client.println("var ctx=document.getElementById('chart');");

client.println("new Chart(ctx,{");

client.println("type:'pie',");

client.println("data:{");

client.println("labels:['Sleeping','Walking','Running'],");

client.println("datasets:[{");

client.println("data:[40,35,25],");

```

```
client.println("backgroundColor:['#4A90E2','2ECC71','#F39C12']");
client.println("}");
client.println("}");
client.println("}");
client.println("</script>");
client.println("</div>");
client.println("</body>");
client.println("</html>");
client.stop();
}
delay(800);
}
```